

1. Write a Python program to generate a complete summary of an employee dataset showing:
 - Number of rows
 - Number of columns
 - Data types
 - Missing values
 - Statistical summary
2. Write a Python program to calculate the frequency distribution of students' grades and display the results using a bar chart.
3. Write a Python program to plot the histogram of house prices and identify whether the data is normally distributed or skewed.
4. Write a Python program to create a box plot of salary data and identify the presence of outliers.
5. Write a Python program to detect outliers in students' marks using the Interquartile Range (IQR) method.
6. Write a Python program to remove outliers from a sales dataset using the IQR method and display the cleaned dataset.
7. Write a Python program to plot histograms and box plots before and after removing outliers from a dataset.
8. Download the Iris or Titanic dataset and perform Exploratory Data Analysis by:
 - Displaying dataset information
 - Finding missing values
 - Computing descriptive statistics
 - Plotting histograms
 - Plotting box plots
 - Detecting outliers
 - Removing outliers
 - Saving the cleaned dataset

Experiment 1 .Employee Dataset Summary using Python

Aim

To read an employee dataset and display the number of rows, columns, data types, missing values, and statistical summary using Pandas.

Algorithm

1. Import the Pandas library.
2. Read the employee dataset using read_csv().
3. Display the number of rows and columns.
4. Display the data types of all columns.
5. Find missing values using isnull().sum().
6. Display statistical summary using describe().
7. End the program.

CODE:

```
import pandas as pd

def employee_summary(file):
    df = pd.read_csv(file)
    print(df)
    print("Number of Rows:", df.shape[0])
    print("Number of Columns:", df.shape[1])

    print("\nData Types:")
    print(df.dtypes)

    print("\nMissing Values:")
    print(df.isnull().sum())

    print("\nStatistical Summary:")
    print(df.describe(include="all"))
employee_summary("Employee.csv")
```

OUTPUT:

```
...      Education  JoiningYear  City  PaymentTier  Age  Gender  EverBenched  \
0      Bachelors      2017      Bangalore      3      34      Male      No
1      Bachelors      2013      Pune      1      28      Female      No
2      Bachelors      2014      New Delhi      3      38      Female      No
3      Masters      2016      Bangalore      3      27      Male      No
4      Masters      2017      Pune      3      24      Male      Yes
...      ...      ...      ...      ...      ...      ...      ...
4648  Bachelors      2013      Bangalore      3      26      Female      No
4649  Masters      2013      Pune      2      37      Male      No
4650  Masters      2018      New Delhi      3      27      Male      No
4651  Bachelors      2012      Bangalore      3      30      Male      Yes
4652  Bachelors      2015      Bangalore      3      33      Male      Yes

      ExperienceInCurrentDomain  LeaveOrNot
0                             0           0
1                             3           1
2                             2           0
3                             5           1
4                             2           1
...                             ...       ...
4648                          4           0
4649                          2           1
4650                          5           1
4651                          2           0
4652                          4           0

[4653 rows x 9 columns]
Number of Rows: 4653
Number of Columns: 9
```

```

Data Types:
Education          object
JoiningYear        int64
City               object
PaymentTier        int64
Age               int64
Gender             object
EverBenched        object
ExperienceInCurrentDomain int64
LeaveOrNot          int64
dtype: object

```

```

Missing Values:
Education          0
JoiningYear        0
City               0
PaymentTier        0
Age               0
Gender             0
EverBenched        0
ExperienceInCurrentDomain 0
LeaveOrNot          0
dtype: int64

```

... Statistical Summary:

	Education	JoiningYear	City	PaymentTier	Age	Gender	\
count	4653	4653.000000	4653	4653.000000	4653.000000	4653	
unique	3	NaN	3	NaN	NaN	2	
top	Bachelors	NaN	Bangalore	NaN	NaN	Male	
freq	3601	NaN	2228	NaN	NaN	2778	
mean	NaN	2015.062970	NaN	2.698259	29.393295	NaN	
std	NaN	1.863377	NaN	0.561435	4.826087	NaN	
min	NaN	2012.000000	NaN	1.000000	22.000000	NaN	
25%	NaN	2013.000000	NaN	3.000000	26.000000	NaN	
50%	NaN	2015.000000	NaN	3.000000	28.000000	NaN	
75%	NaN	2017.000000	NaN	3.000000	32.000000	NaN	
max	NaN	2018.000000	NaN	3.000000	41.000000	NaN	

	EverBenched	ExperienceInCurrentDomain	LeaveOrNot
count	4653	4653.000000	4653.000000
unique	2	NaN	NaN
top	No	NaN	NaN
freq	4175	NaN	NaN
mean	NaN	2.905652	0.343864
std	NaN	1.558240	0.475047
min	NaN	0.000000	0.000000
25%	NaN	2.000000	0.000000
50%	NaN	3.000000	0.000000
75%	NaN	4.000000	1.000000
max	NaN	7.000000	1.000000

Experiment 2: Frequency Distribution of Students' Grades

Aim

To calculate the frequency of students' grades and visualize it using a bar chart.

Algorithm

1. Import Pandas and Matplotlib.
2. Read the grades dataset.
3. Count the frequency of each grade using `value_counts()`.
4. Display the frequency table.
5. Plot a bar chart.
6. Add title and axis labels.
7. Display the chart.

CODE:

Q2

```
import pandas as pd
import matplotlib.pyplot as plt

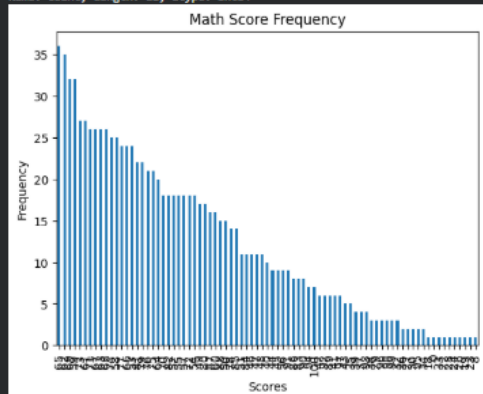
def grade_frequency(file):
    df = pd.read_csv(file)
    print("Available columns:", df.columns)
    freq = df["math score"].value_counts()
    print(freq)

    freq.plot(kind="bar")
    plt.title("Math Score Frequency")
    plt.xlabel("Scores")
    plt.ylabel("Frequency")
    plt.show()

grade_frequency("StudentsPerformance.csv")
```

OUTPUT:

```
grade_frequency("StudentsPerformance.csv")
--- Available columns: Index(['gender', 'race/ethnicity', 'parental level of education', 'lunch',
                             'test preparation course', 'math score', 'reading score',
                             'writing score'],
                             dtype='object')
math score
65    36
62    35
69    32
59    32
73    27
..
24     1
26     1
19     1
23     1
8       1
Name: count, Length: 81, dtype: int64
```



Experiment 3: Histogram of House Prices

Aim

To plot a histogram of house prices and identify whether the data is normally distributed or skewed.

Algorithm

1. Import Pandas and Matplotlib.
2. Read the house price dataset.
3. Select the price column.
4. Plot a histogram using plt.hist().
5. Add title and axis labels.
6. Display the histogram.
7. Observe the distribution.

CODE:

```
Q3

import pandas as pd
import matplotlib.pyplot as plt

def house_price_histogram(file):
    df = pd.read_csv(file)

    print("Available columns:", df.columns)
    plt.hist(df["price"], bins=10)
    plt.title("House Prices")
    plt.xlabel("Price")
    plt.ylabel("Frequency")
    plt.show()

    print("Check the histogram shape to determine whether the data is normally distributed or skewed.")

house_price_histogram("kc_house_data.csv")
```

OUTPUT:



Experiment 4: Box Plot of Salary Data

Title

Box Plot of Salary Data

Aim

To create a box plot for salary data and identify the presence of outliers.

Algorithm

1. Import Pandas and Matplotlib.
2. Read the salary dataset.
3. Select the salary column.
4. Plot a box plot using plt.boxplot().
5. Add title and labels.
6. Display the plot.
7. Identify outliers outside the whiskers.

CODE:

Q4

```
import pandas as pd
import matplotlib.pyplot as plt

def salary_boxplot(file):
    df = pd.read_csv(file)

    print("Available columns:", df.columns)

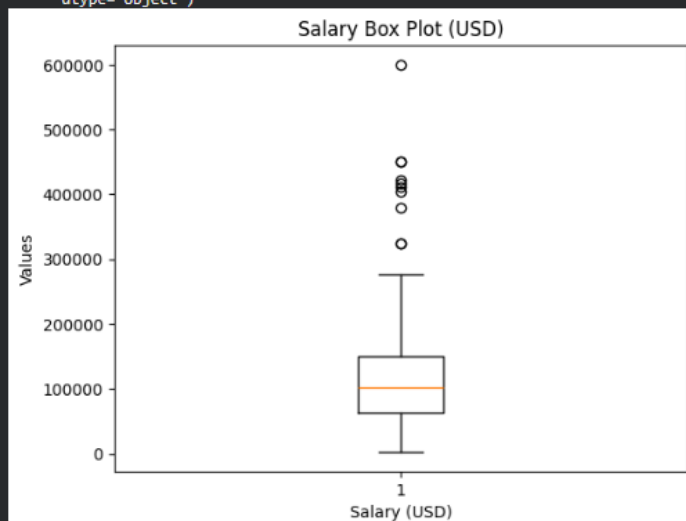
    plt.boxplot(df["salary_in_usd"])
    plt.title("Salary Box Plot (USD)")
    plt.xlabel("Salary (USD)")
    plt.ylabel("Values")
    plt.show()

    print("Points outside the whiskers are outliers.")

salary_boxplot("ds_salaries.csv")
```

OUTPUT:

```
*** Available columns: Index(['Unnamed: 0', 'work_year', 'experience_level', 'employment_type',
                             'job_title', 'salary', 'salary_currency', 'salary_in_usd',
                             'employee_residence', 'remote_ratio', 'company_location',
                             'company_size'],
                             dtype='object')
```



Points outside the whiskers are outliers.

Experiment 5: Detect Outliers using IQR

Title

Outlier Detection using Interquartile Range (IQR)

Aim

To detect outliers in students' marks using the Interquartile Range (IQR) method.

Algorithm

1. Import Pandas.
2. Read the students' marks dataset.
3. Calculate Q1 and Q3.
4. Calculate $IQR = Q3 - Q1$.
5. Find lower and upper limits.
6. Detect values outside these limits.
7. Display the outliers.

CODE:

Q5

```
import pandas as pd

def detect_outliers(file):
    df = pd.read_csv(file)

    print("Available columns:", df.columns)
    Q1 = df["math score"].quantile(0.25)
    Q3 = df["math score"].quantile(0.75)

    IQR = Q3 - Q1

    lower = Q1 - 1.5 * IQR
    upper = Q3 + 1.5 * IQR

    outliers = df[(df["math score"] < lower) | (df["math score"] > upper)]

    print(outliers)

detect_outliers("StudentsPerformance.csv")
```

OUTPUT:

```
Available columns: Index(['gender', 'race/ethnicity', 'parental level of education', 'lunch',
                          'test preparation course', 'math score', 'reading score',
                          'writing score'],
                          dtype='object')
gender race/ethnicity parental level of education lunch \
17 female group B some high school free/reduced
59 female group C some high school free/reduced
145 female group C some college free/reduced
338 female group B some high school free/reduced
466 female group D associate's degree free/reduced
787 female group B some college standard
842 female group B high school free/reduced
980 female group B high school free/reduced

test preparation course math score reading score writing score
17 none 18 32 28
59 none 0 17 10
145 none 22 39 33
338 none 24 38 27
466 none 26 31 38
787 none 19 38 32
842 completed 23 44 36
980 none 8 24 23
```

Experiment 6: Remove Outliers using IQR

Title

Removing Outliers using IQR Method

Aim

To remove outliers from a sales dataset using the Interquartile Range (IQR) method.

Algorithm

1. Import Pandas.
2. Read the sales dataset.
3. Calculate Q1 and Q3.
4. Calculate the IQR.
5. Find the lower and upper limits.
6. Remove values outside the limits.
7. Display the cleaned dataset.

CODE:

```
Q6

import pandas as pd

def remove_outliers(file):
    df = pd.read_csv(file, encoding='latin1')

    Q1 = df["Sales"].quantile(0.25)
    Q3 = df["Sales"].quantile(0.75)

    IQR = Q3 - Q1

    lower = Q1 - 1.5 * IQR
    upper = Q3 + 1.5 * IQR

    clean = df[(df["Sales"] >= lower) & (df["Sales"] <= upper)]

    print(clean)

remove_outliers("Sample - Superstore.csv")
```

OUTPUT:

```
remove_outliers("Sample - Superstore.csv")

...
9990 9991 CA-2017-121258 2/26/2017 3/3/2017 Standard Class
9991 9992 CA-2017-121258 2/26/2017 3/3/2017 Standard Class
9992 9993 CA-2017-121258 2/26/2017 3/3/2017 Standard Class
9993 9994 CA-2017-119914 5/4/2017 5/9/2017 Second Class

...
Customer ID Customer Name Segment Country City \
0 CG-12520 Claire Gute Consumer United States Henderson
2 DV-13045 Darrin Van Huff Corporate United States Los Angeles
4 SO-20335 Sean O'Donnell Consumer United States Fort Lauderdale
5 BH-11710 Brosina Hoffman Consumer United States Los Angeles
6 BH-11710 Brosina Hoffman Consumer United States Los Angeles
...
...
9989 TB-21400 Tom Boeckenhauer Consumer United States Miami
9990 DB-13060 Dave Brooks Consumer United States Costa Mesa
9991 DB-13060 Dave Brooks Consumer United States Costa Mesa
9992 DB-13060 Dave Brooks Consumer United States Costa Mesa
9993 CC-12220 Chris Cortes Consumer United States Westminster

...
Postal Code Region Product ID Category Sub-Category \
0 ... 42420 South FUR-BO-10001708 Furniture Bookcases
2 ... 90036 West OFF-LA-10000240 Office Supplies Labels
4 ... 33311 South OFF-ST-10000760 Office Supplies Storage
5 ... 90032 West FUR-FU-10001487 Furniture Furnishings
6 ... 90032 West OFF-AR-10002833 Office Supplies Art
...
...
9989 ... 33180 South FUR-FU-10001889 Furniture Furnishings
9990 ... 92627 West FUR-FU-10000747 Furniture Furnishings
9991 ... 92627 West TEC-PH-10003645 Technology Phones
9992 ... 92627 West OFF-PA-10004041 Office Supplies Paper
9993 ... 92683 West OFF-AP-10002684 Office Supplies Appliances

...
Product Name Sales Quantity \
0 Bush Somerset Collection Bookcase 261.960 2
2 Self-Adhesive Address Labels for Typewriters b... 14.620 2
4 Eldon Fold 'N Roll Cart System 22.368 2
5 Eldon Expressions Mood and Plastic Desk Access... 48.860 7
6 Newell 322 7.280 4
...
...
9989 Ultra Door Pull Handle 25.248 3
9990 Temex BI-RE Series Chair Mats for Low Pile Car... 91.960 2
9991 Aastra 571 VoIP phone 258.576 2
9992 It's Hot Message Books with Stickers, 2 3/4" x 5" 29.600 4
9993 Acco 7-Outlet Masterpiece Power Center, Whitou... 243.160 2

Discount Profit
0 0.0 41.9136
2 0.0 6.8714
4 0.2 2.5164
5 0.0 14.1694
6 0.0 1.9656
...
...
9989 0.2 4.1028
9990 0.0 15.6332
9991 0.2 19.3932
9992 0.0 13.3200
9993 0.0 72.9480
```


Experiment 7: Histograms and Box Plots Before and After Removing Outliers

Title

Comparison of Data Before and After Removing Outliers

Aim

To visualize the dataset before and after removing outliers using histograms and box plots.

Algorithm

1. Import Pandas and Matplotlib.
2. Read the dataset.
3. Plot histogram and box plot before removing outliers.
4. Calculate Q1, Q3, and IQR.
5. Remove outliers.
6. Plot histogram after removing outliers.
7. Plot box plot after removing outliers.
8. Compare the results.

CODE:

Q7

```
import pandas as pd
import matplotlib.pyplot as plt

def before_after(file):
    df = pd.read_csv(file, encoding='latin1')

    plt.hist(df["Sales"])
    plt.title("Before Removing Outliers")
    plt.show()

    plt.boxplot(df["Sales"])
    plt.title("Before Box Plot")
    plt.show()

    Q1 = df["Sales"].quantile(0.25)
    Q3 = df["Sales"].quantile(0.75)

    IQR = Q3 - Q1

    lower = Q1 - 1.5 * IQR
    upper = Q3 + 1.5 * IQR

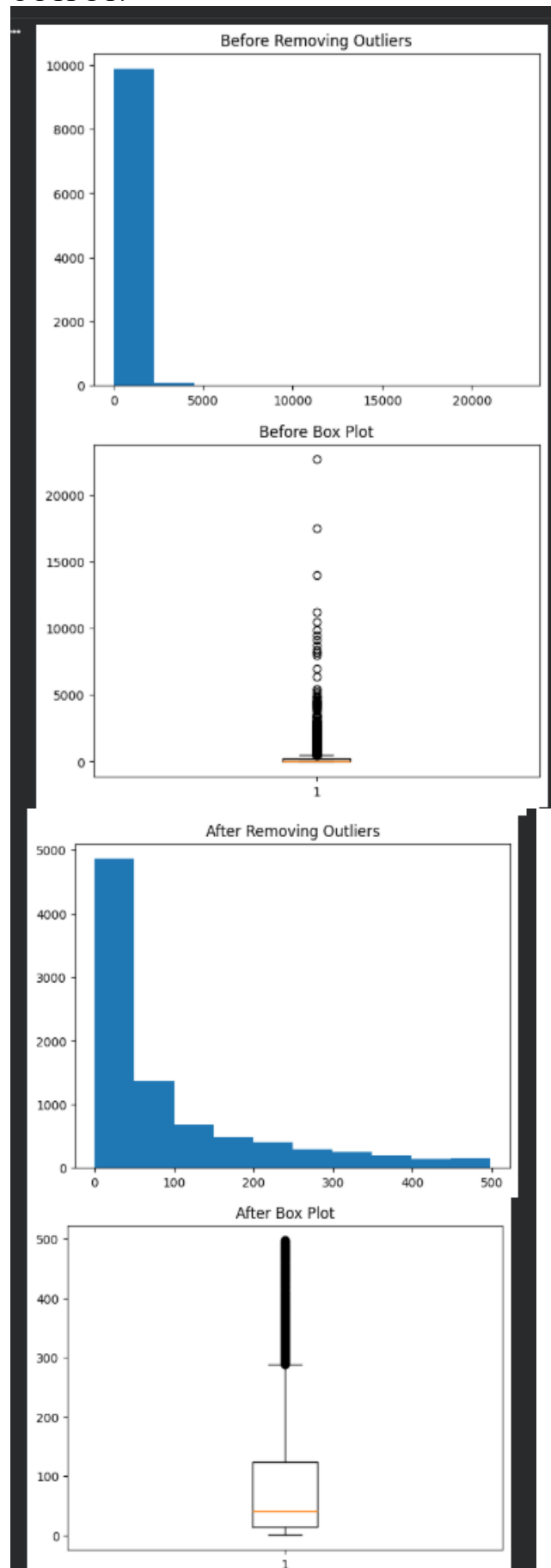
    clean = df[(df["Sales"] >= lower) & (df["Sales"] <= upper)]

    plt.hist(clean["Sales"])
    plt.title("After Removing Outliers")
    plt.show()

    plt.boxplot(clean["Sales"])
    plt.title("After Box Plot")
    plt.show()

before_after("Sample - Superstore.csv")
```

OUTPUT:



Experiment 8: Exploratory Data Analysis (EDA) on Iris/Titanic Dataset

Aim

To perform Exploratory Data Analysis on the Iris or Titanic dataset by displaying dataset information, finding missing values, computing descriptive statistics, plotting histograms and box plots, detecting and removing outliers, and saving the cleaned dataset.

Algorithm

1. Import Pandas and Matplotlib.
2. Read the Iris or Titanic dataset.
3. Display dataset information using info().
4. Find missing values using isnull().sum().
5. Display descriptive statistics using describe().
6. Plot histograms for numerical columns.
7. Plot box plots for numerical columns.
8. Detect outliers using the IQR method.
9. Remove outliers from the dataset.
10. Save the cleaned dataset as a new CSV file.
11. End the program.

CODE:

```
Q8

import pandas as pd
import matplotlib.pyplot as plt

def eda(file):
    df = pd.read_csv(file)

    print(df.info())

    print("\nMissing Values")
    print(df.isnull().sum())

    print("\nStatistics")
    print(df.describe())

    df.hist(figsize=(10,8))
    plt.show()

    df.boxplot(figsize=(10,6))
    plt.show()

    num = df.select_dtypes(include="number")

    Q1 = num.quantile(0.25)
    Q3 = num.quantile(0.75)

    IQR = Q3 - Q1

    clean = df[~((num < (Q1-1.5*IQR)) | (num > (Q3+1.5*IQR))).any(axis=1)]

    clean.to_csv("Cleaned_dataset.csv", index=False)

    print("Cleaned dataset saved.")

eda("gender_submission.csv")
```

OUTPUT:

